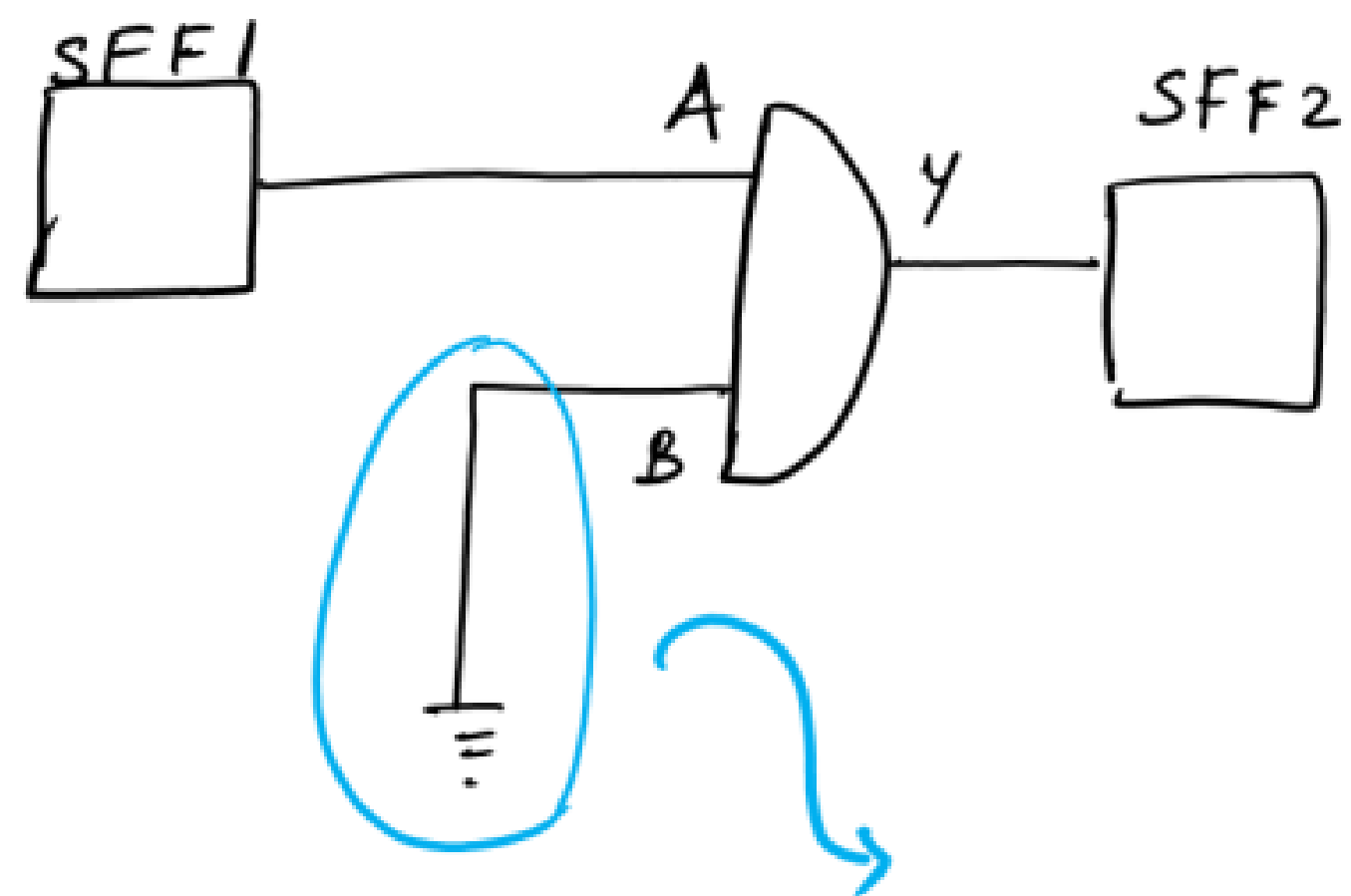


ATPG part 2

EXAMPLE 2:

INTENDED DESIGN:

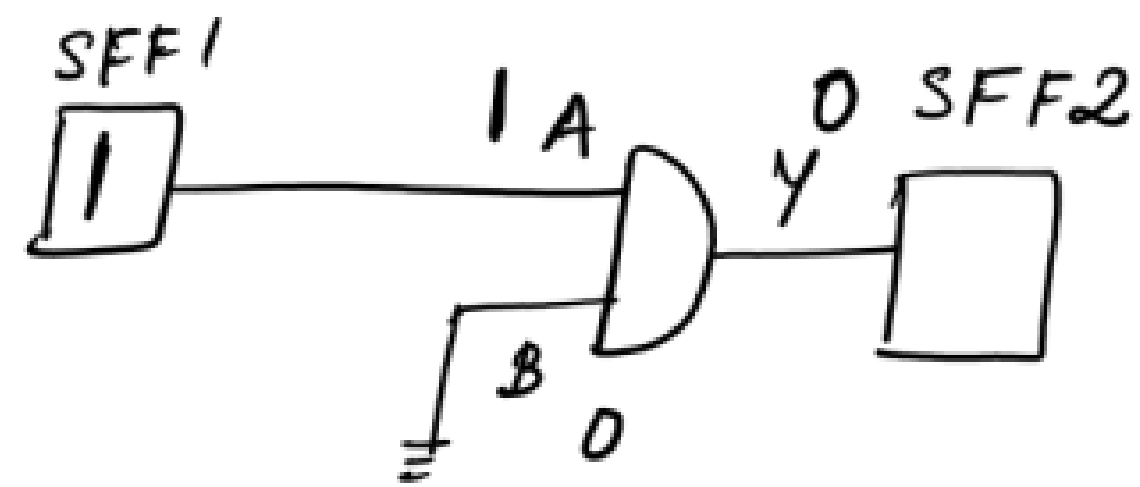


This is intentionally tied to Gnd by the designer to meet the functionality.

Which faults will get detected & which faults will not get detected?

(i) ASAO:-

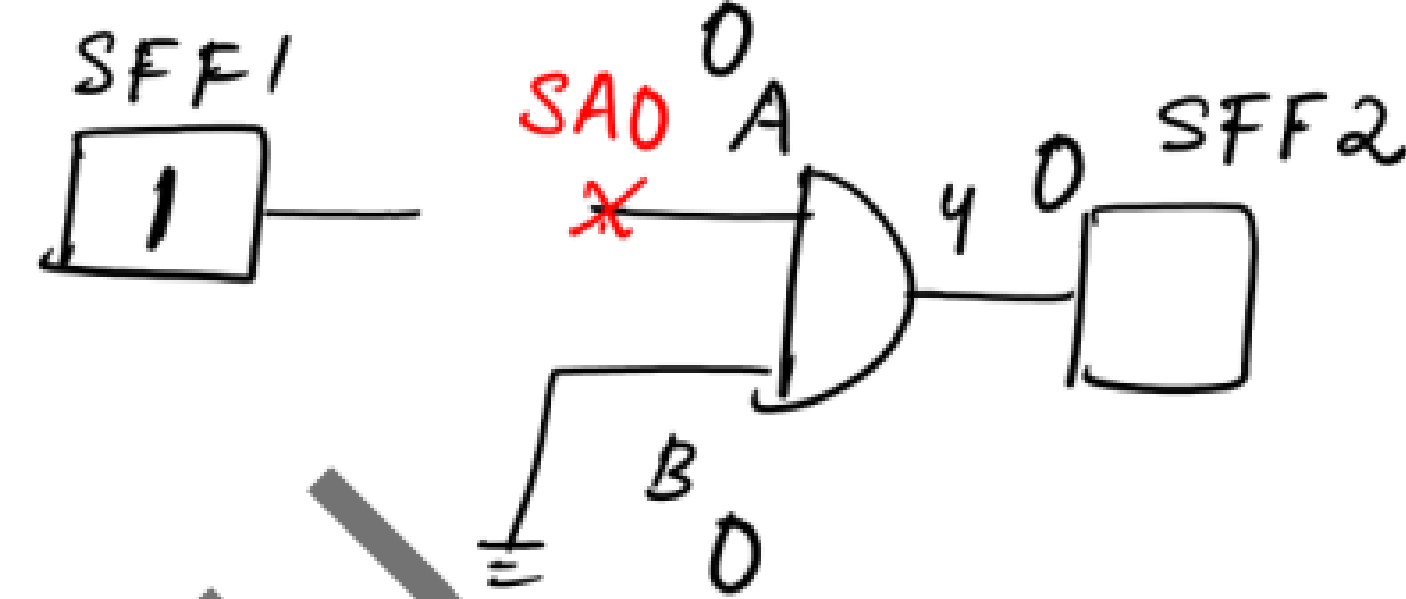
Good m/c



To propagate the value of A net to the obs point, B has to be made 1.

But in this case, tool can't bring 1 at the B net as it is connected to gnd in the design.

Faulty m/c



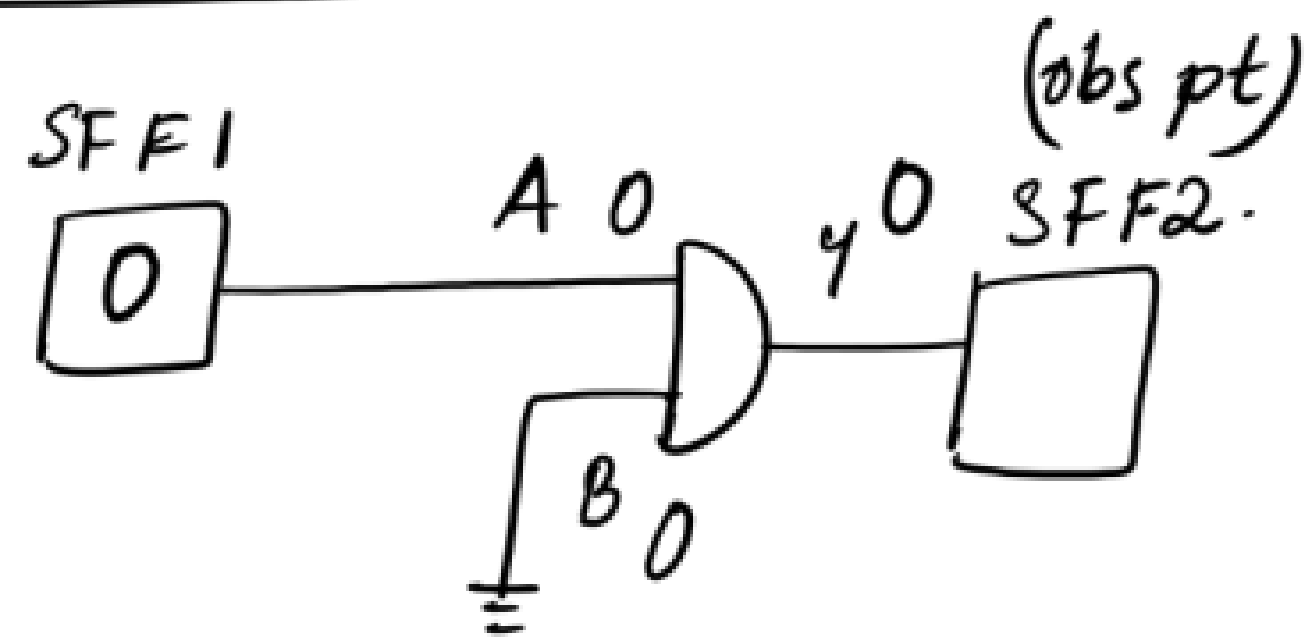
Good value = 0
Faulty value = 0

Good value = Faulty value
∴ ASAO is not detected during fault sim

implies that $\Rightarrow$ if this faulty behaviour occurs after man, it will not get detected

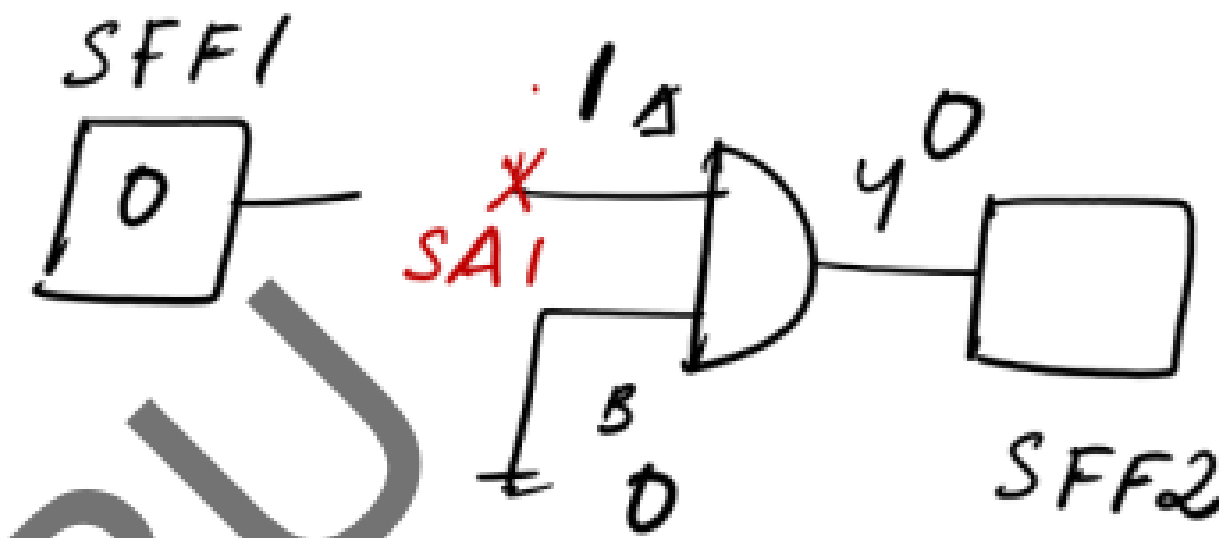
(ii) A SAI

Good m/c



To propagate the value at A net (fault site) to the obs pt, B has to be made 1. But in this case, tool can't bring 1 at the B net because it is connected to gnd in this design.

Faulty m/c (tool has induced the A SAI fault)

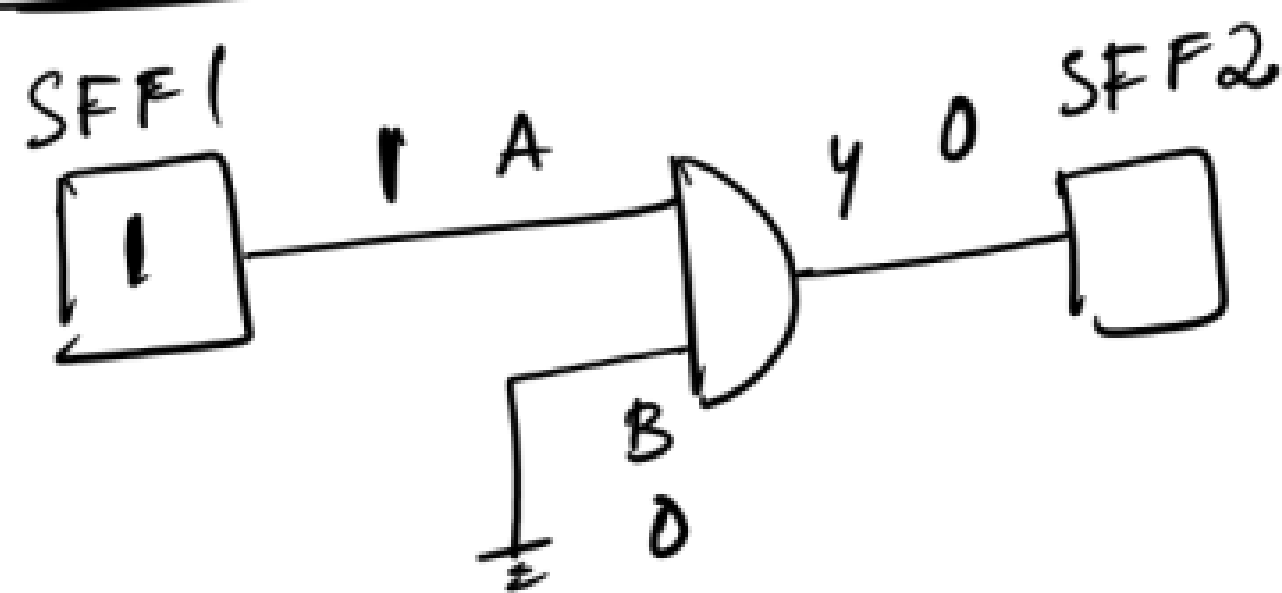


$\therefore$ A SAI is not detected during Fault sim

Good value = 0
Faulty value = 0
Good value = Faulty value
 $\Rightarrow$ if this faulty beh occurs after man, it will not get detected.

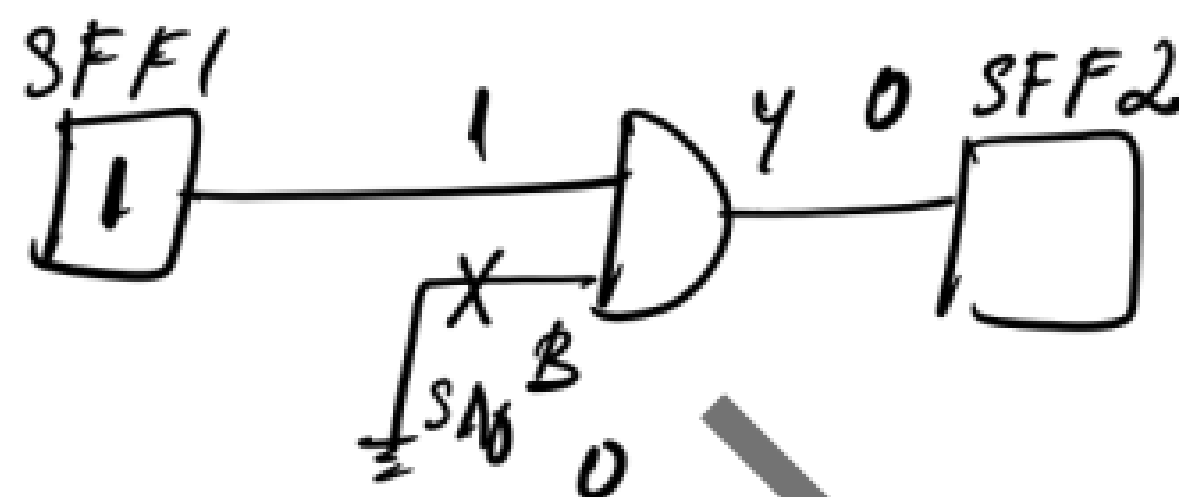
(iii) BSAD :-

Good m/c



Actually to detect BSAD, the tool will try to make $B=1$. But the tool is not able to do it as B is tied to gnd by the design.

Faulty m/c



Good value = 0
Faulty value = 0

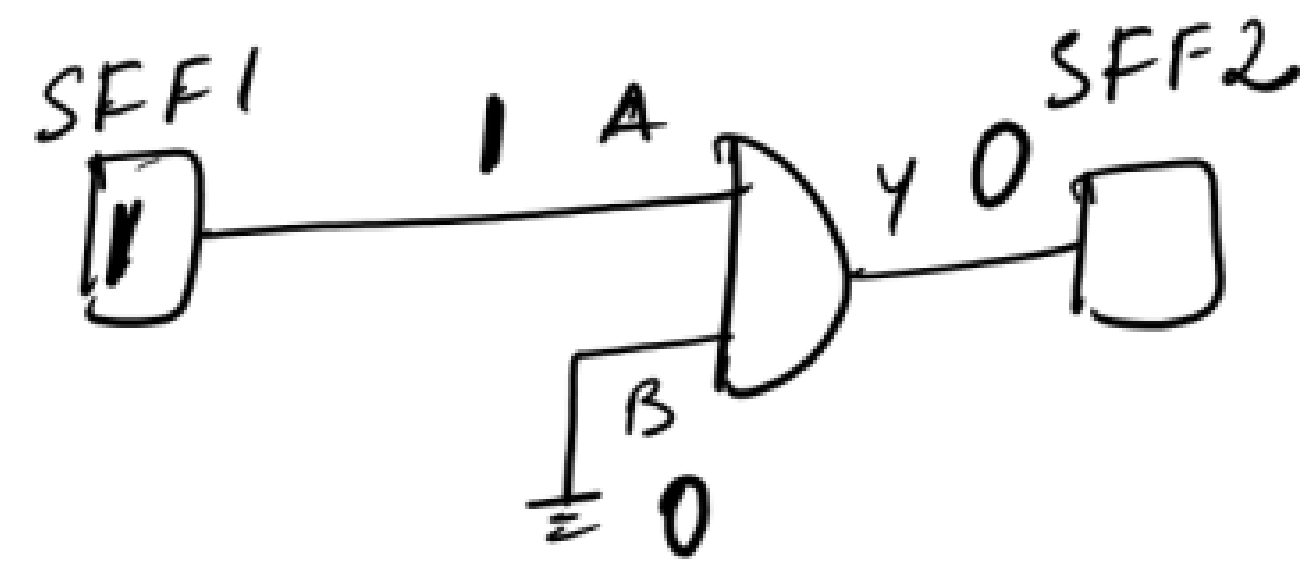
$\therefore$ BSAD is not detected during fault sim

Good value = Faulty value

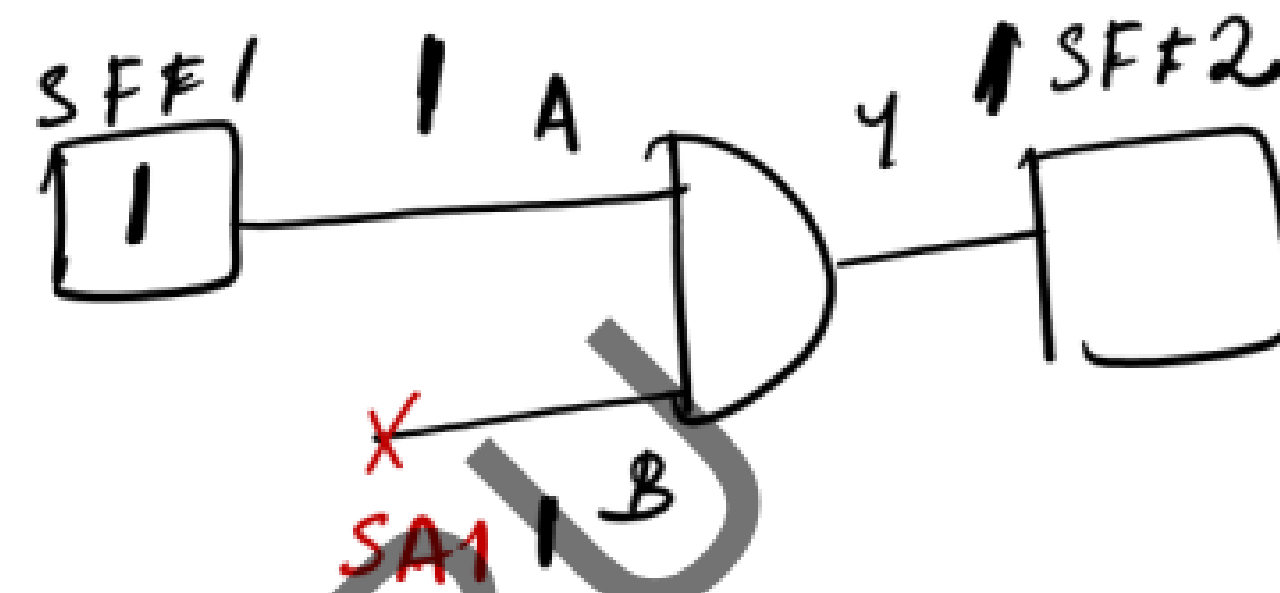
$\hookrightarrow$ If suppose this faulty beh occurs after man, it will not be detected

(iv) BSA1:

Good m/c



Faulty m/c :-



Good value = 0
Faulty value = 1

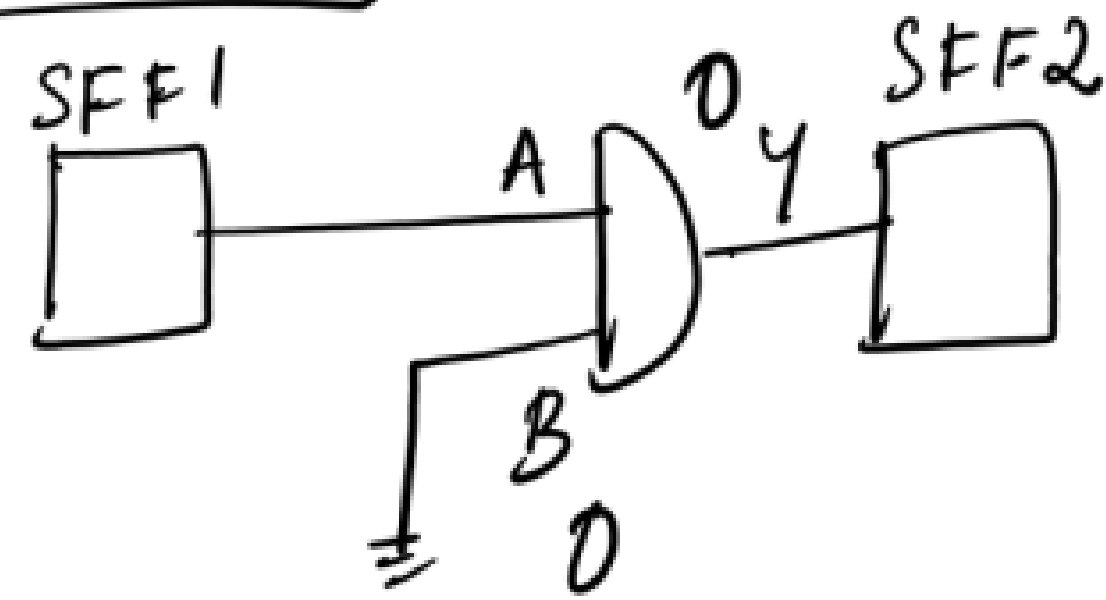
Good value $\neq$ Faulty value
 $\therefore$ BSA1 is detected during fault sim

||> If suppose this faulty beh occurs after

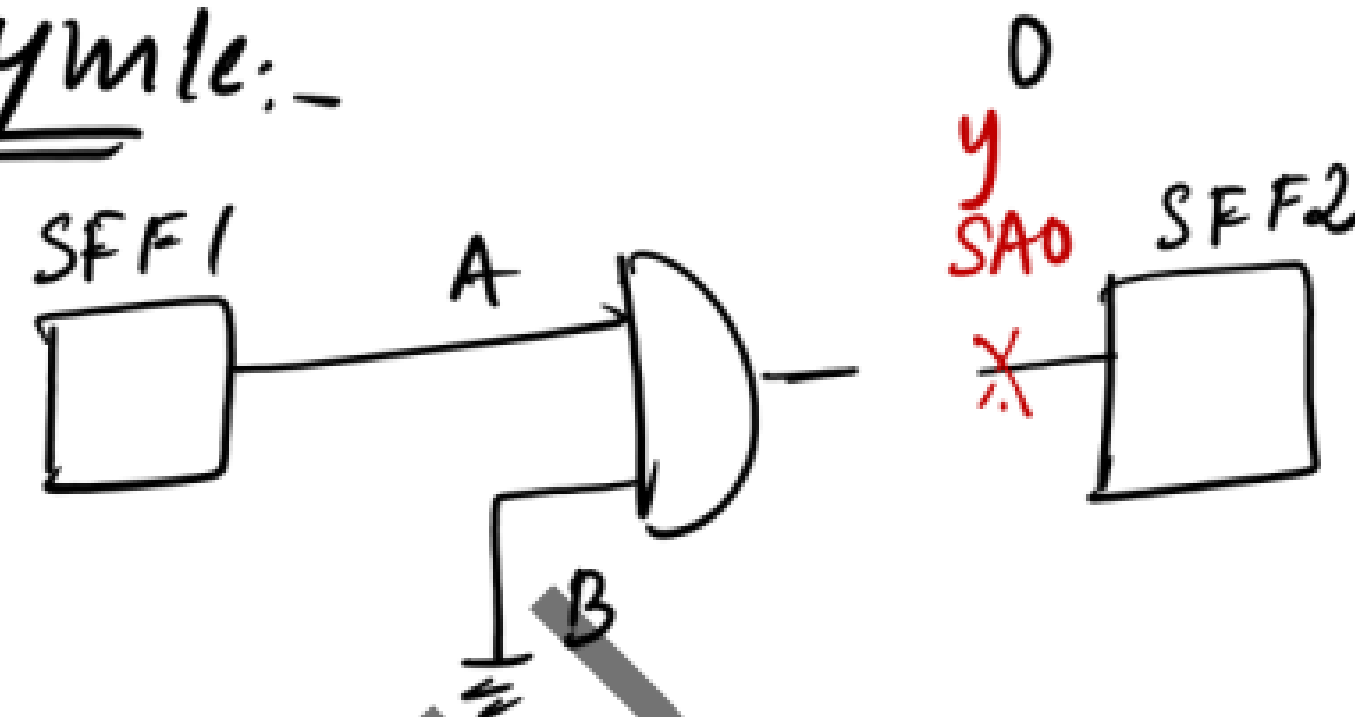
man, it will get detected

(V) YSAO :-

Good m/c :-



Faulty m/c :-



$\therefore$ YSAO is not detected during fault sim

Good value = 0

Faulty value = 0

} both are same

$\hookrightarrow$ If suppose this faulty beh occurs

after man, it will not get

detected

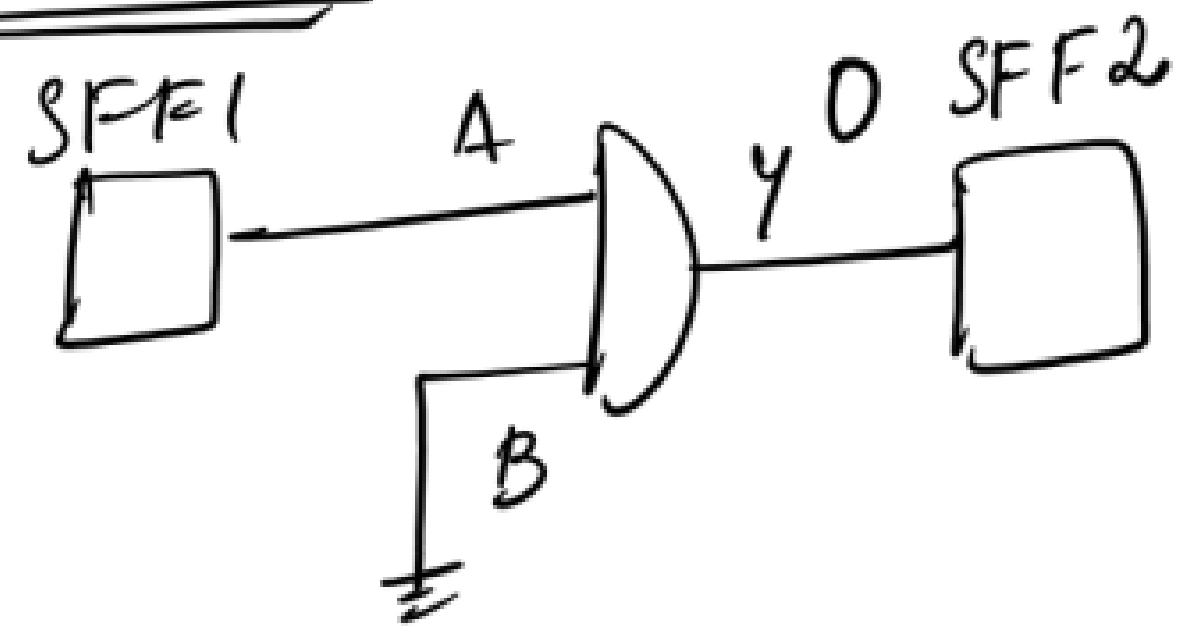
In order to detect YSAO, the tool will try to bring '1' at Y.

Only $A=1$ and $B=1$, tool can bring '1' at Y.

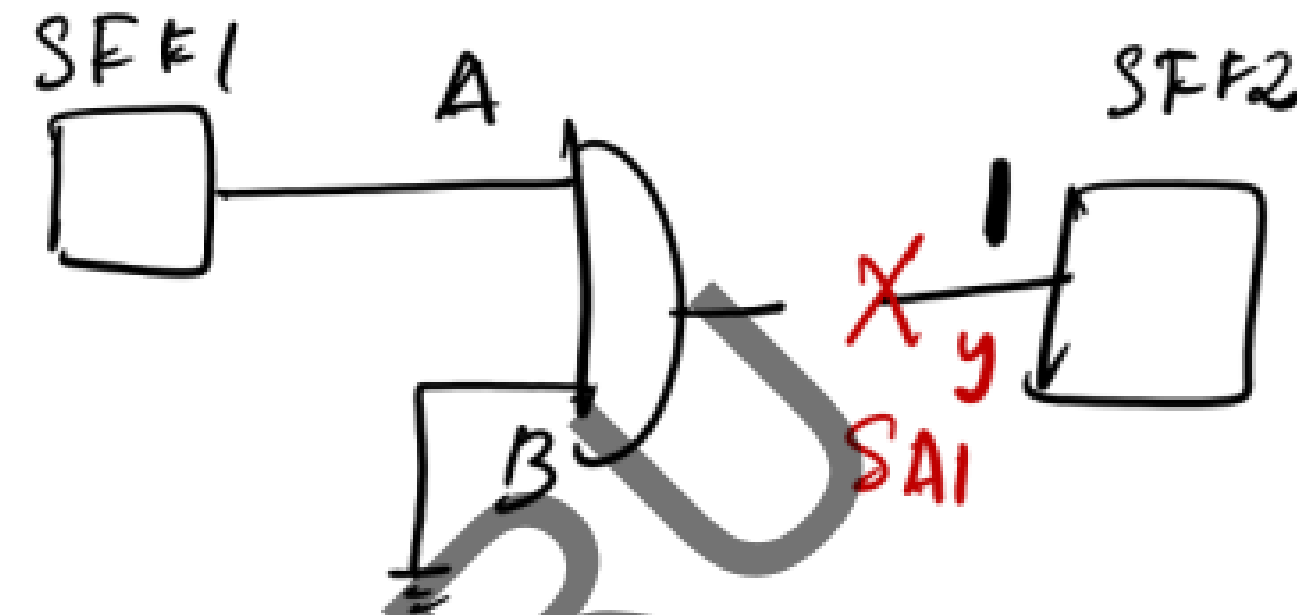
But tool can't bring 1 at B as B is connected to gnd by the design itself. Hence tool can't bring '1' at Y also.

(vi) YSAI :-

Good m/c :-



Faulty m/c :-



Good value = 0

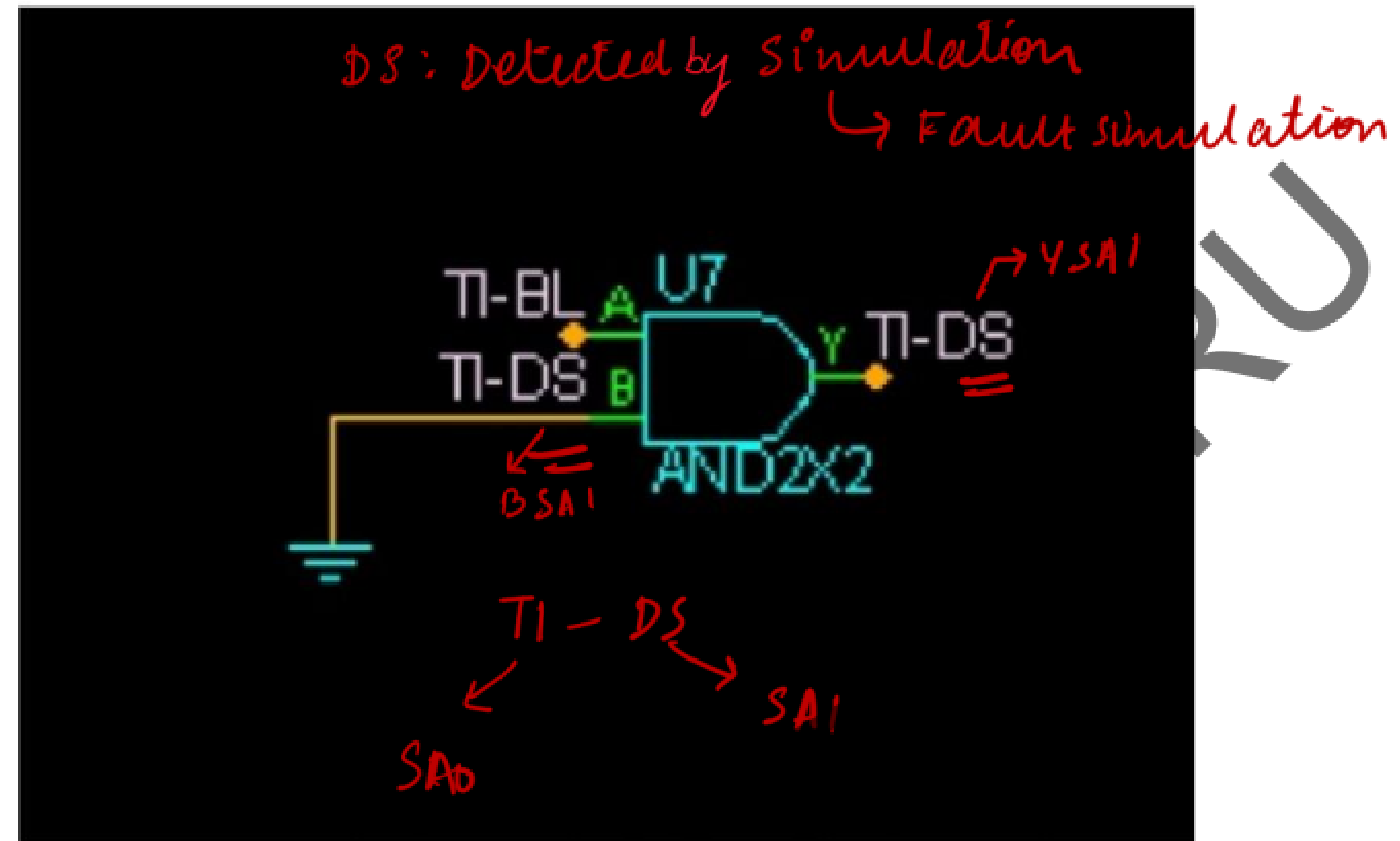
Faulty value = 1

Good value $\neq$ Faulty value

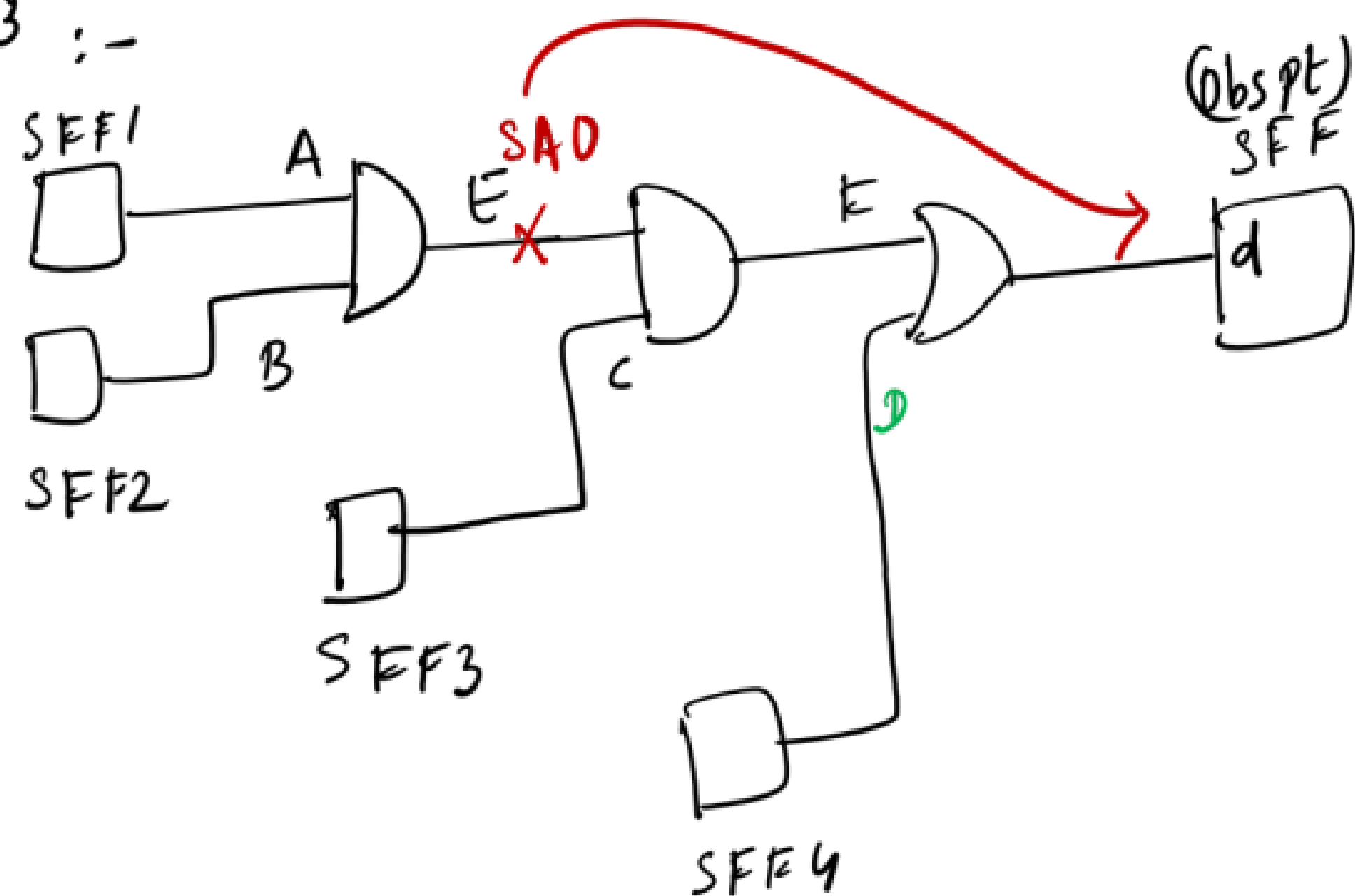
$\therefore$ YSAI is detected during fault sim

$\hookrightarrow$ If suppose this faulty beh occurs after man, it will get detected

BSA1 and YSA1 will get detected



Eg 3 :-



Which pattern will detect ESAO fault?

D algo :-

• Assume a fault $\Rightarrow$ ESAO

• Identify opposite value $\Rightarrow E \leftarrow 1$

• Apply i/p seq to get the } $\Rightarrow \begin{matrix} A = 1 \\ B = 1 \end{matrix}$
opposite value

• propagate the value at the fault site to the obs pt } $\Rightarrow \begin{matrix} C = 1 \\ D = 0 \end{matrix}$

Pattern :- $\begin{matrix} A = 1 & C = 1 \\ B = 1 & D = 0 \end{matrix}$

T.T of AND Gate

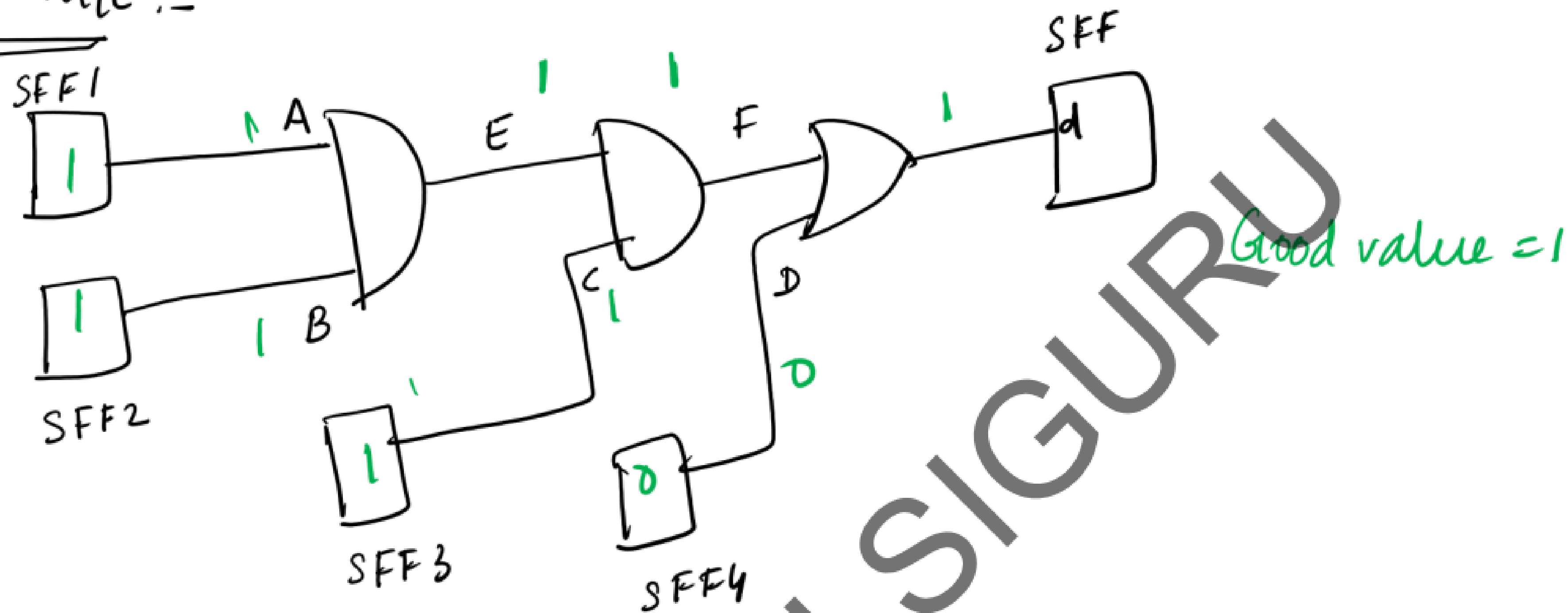
i_1 (E)	i_2 (C)	O/P (F)
0	0	0
0	1	0
1	0	0
1	1	1

T.T of OR gate

i_1 (F)	i_2 (D)	O/P
0	0	0
0	1	1
1	0	1
1	1	1

Fault Simulation:-

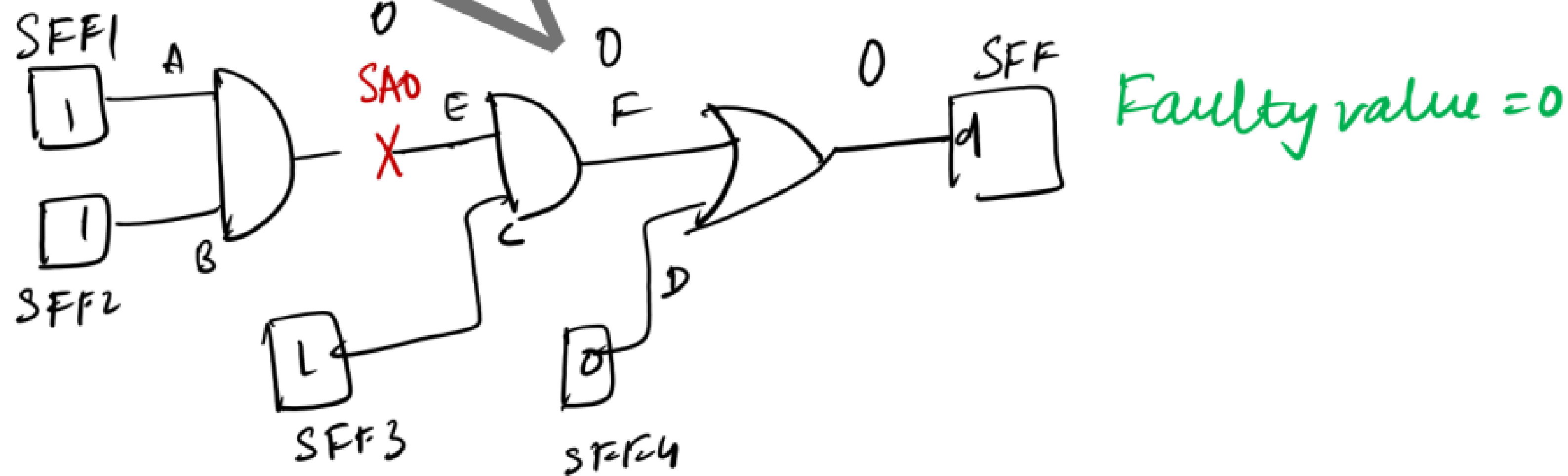
Good m/c :-



Good value $\neq$ Faulty value

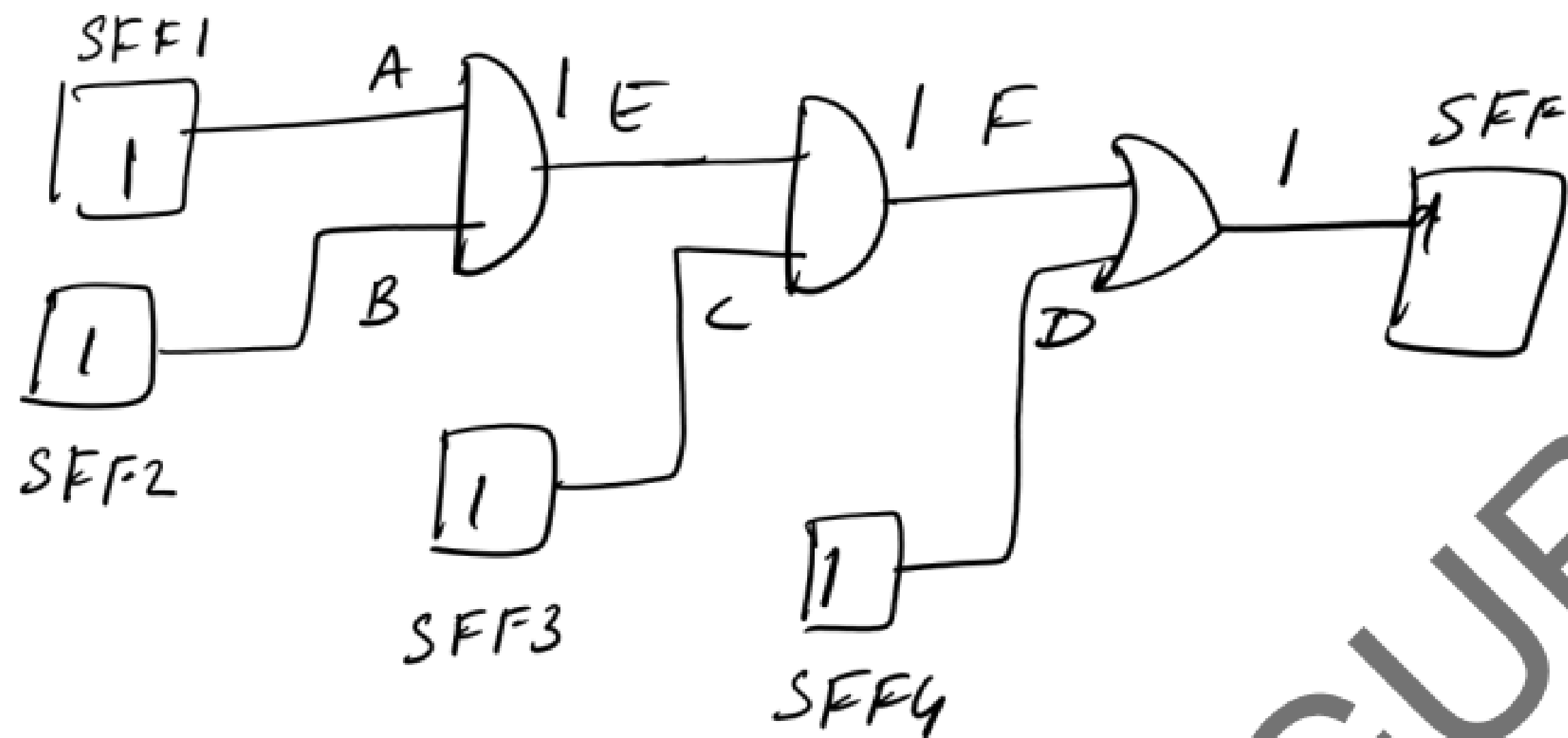
ESAO is detected during fault sim
implies that $\Rightarrow$ if this faulty beh occurs after manufacturing due to the presence of a defect, it will get detected with this pattern

Faulty m/c :-



Will 1111 pattern be able to detect this fault?

Good m/c :-

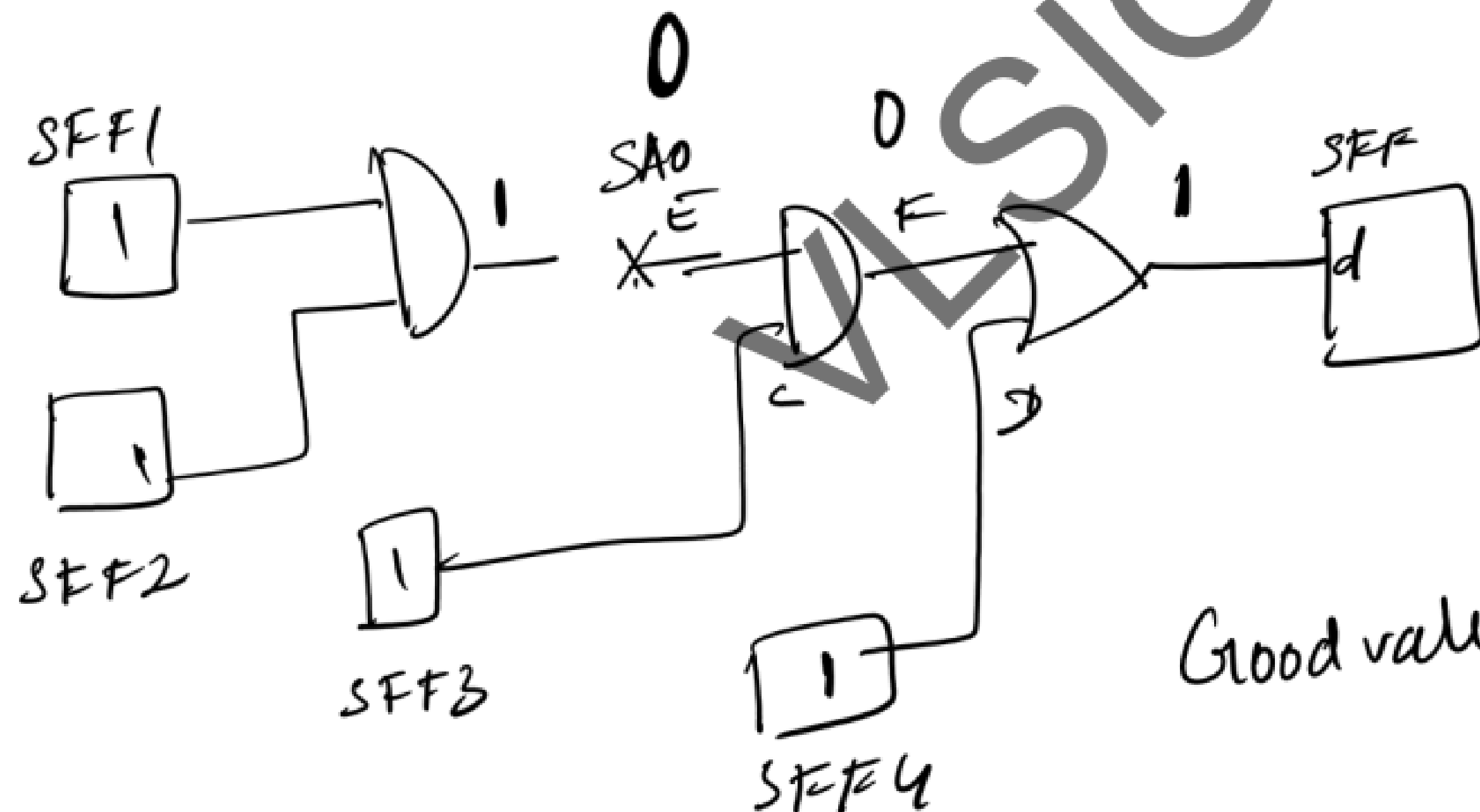


Good value = 1

For this pattern,
Good value = Faulty value.

$\therefore$ ESAO is not detected by this pattern during fault sim.

Faulty m/c :-



Faulty value = 1

Good value = Faulty value = 1 (same)

$\Rightarrow$ if this faulty beh occurs after manufacturing due to the presence of a defect, it will not get detected with this pattern

EXAMPLE 4 :-

$E \rightarrow$ fanout stem

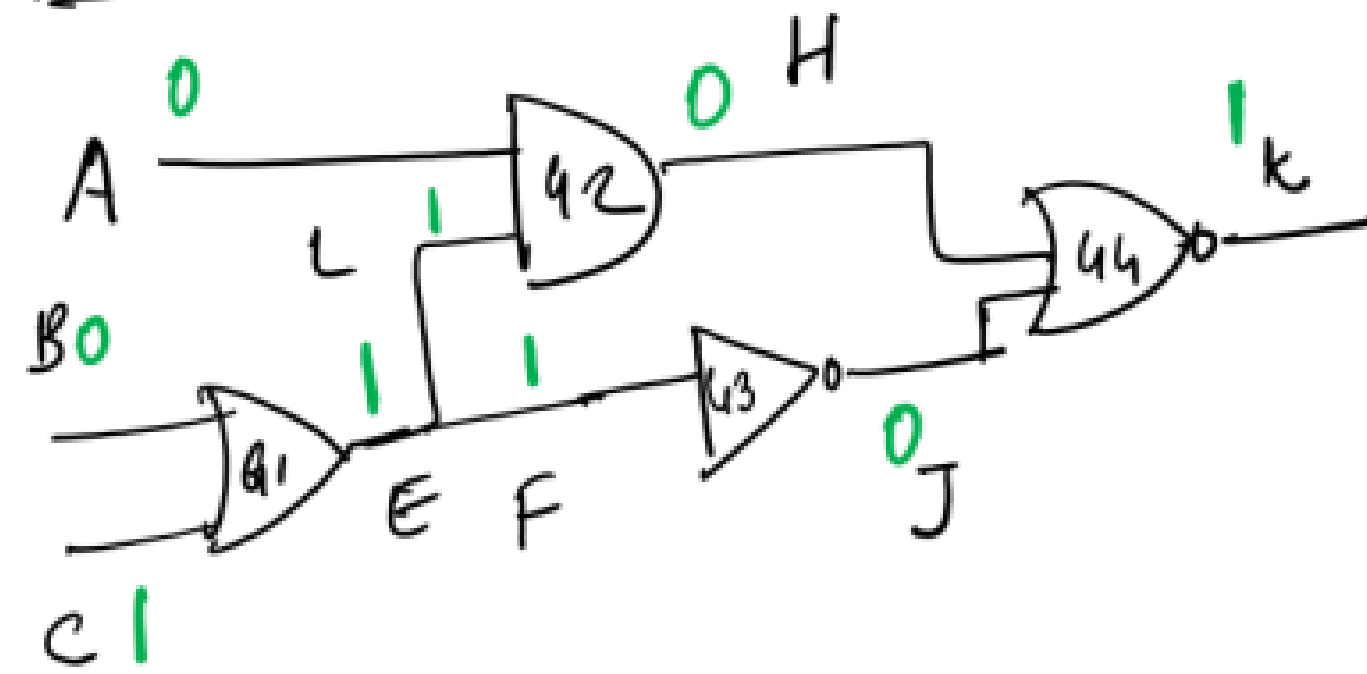
$L, F \rightarrow$ fanout branches

Stuck at fault on fanout stem is not equivalent to Stuck At fault on fanout branches

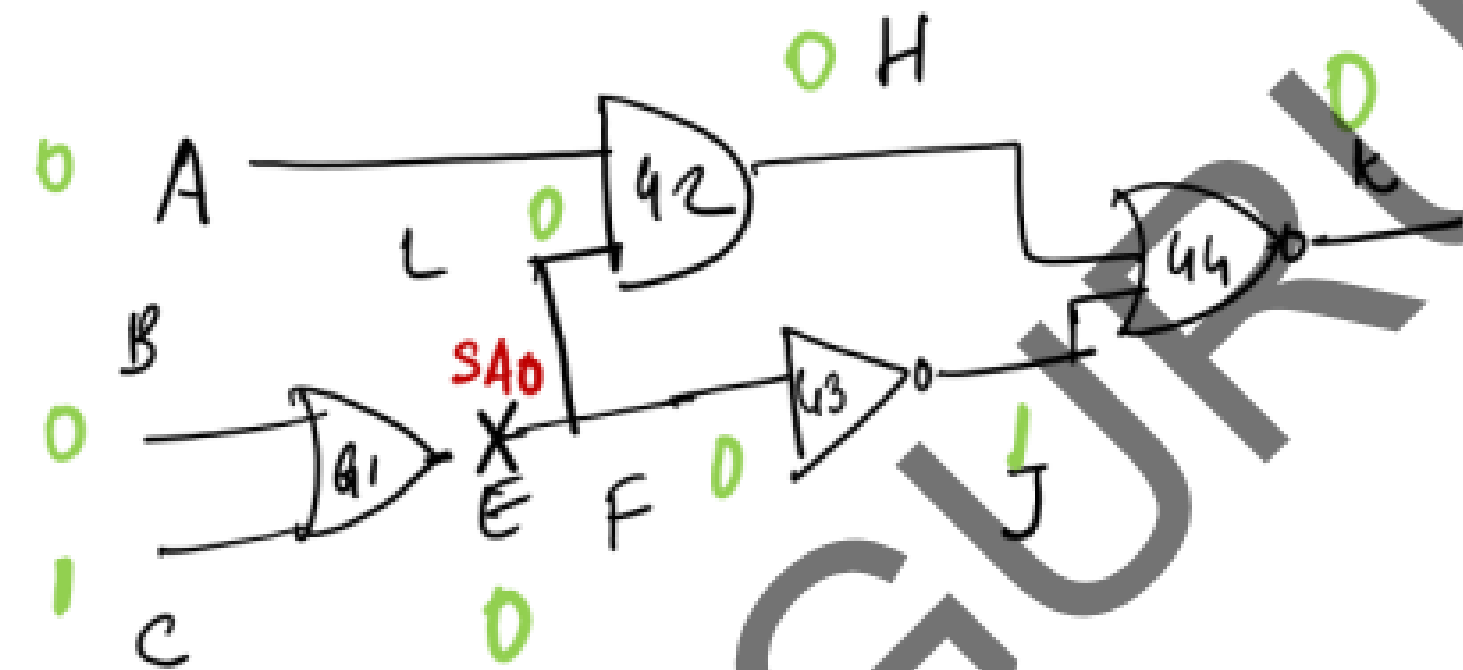
$A, B, C = 001$

Good value = 1

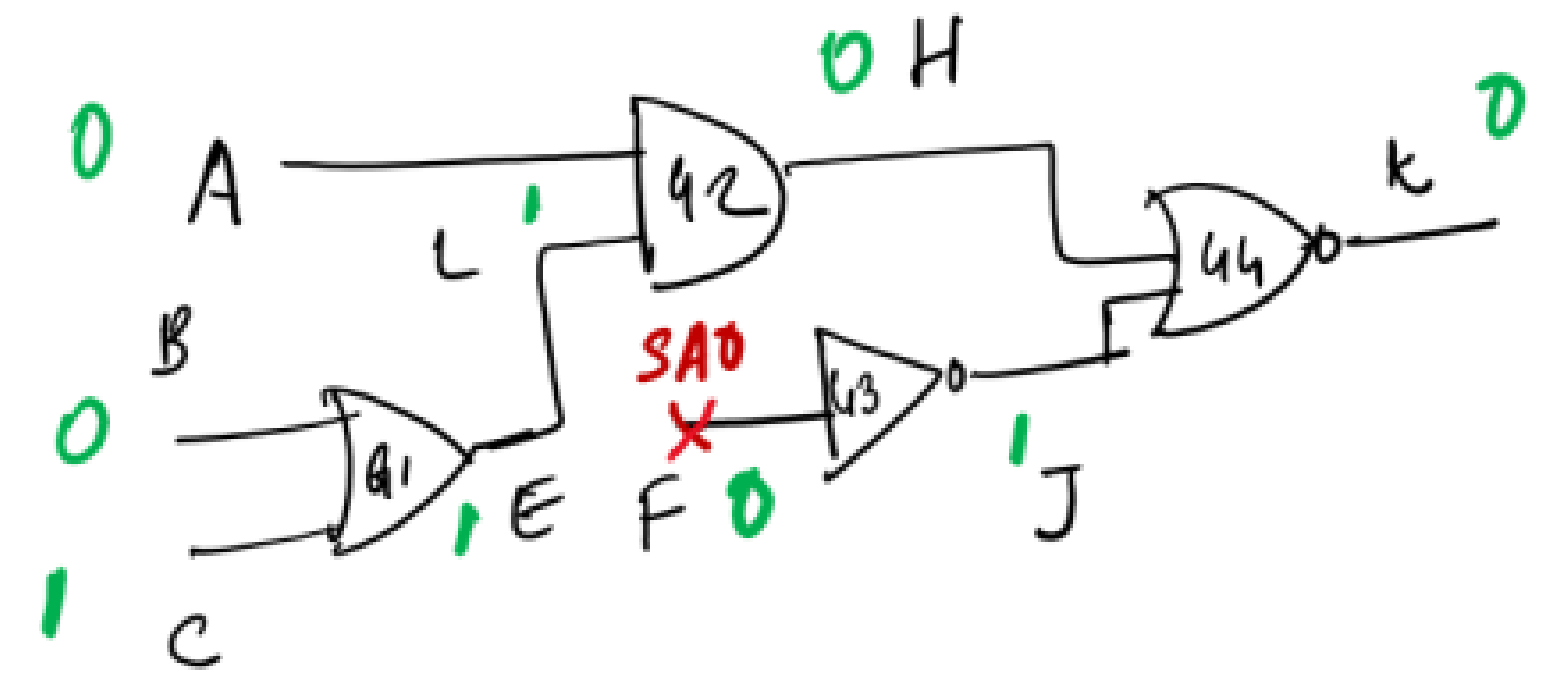
Good m/c



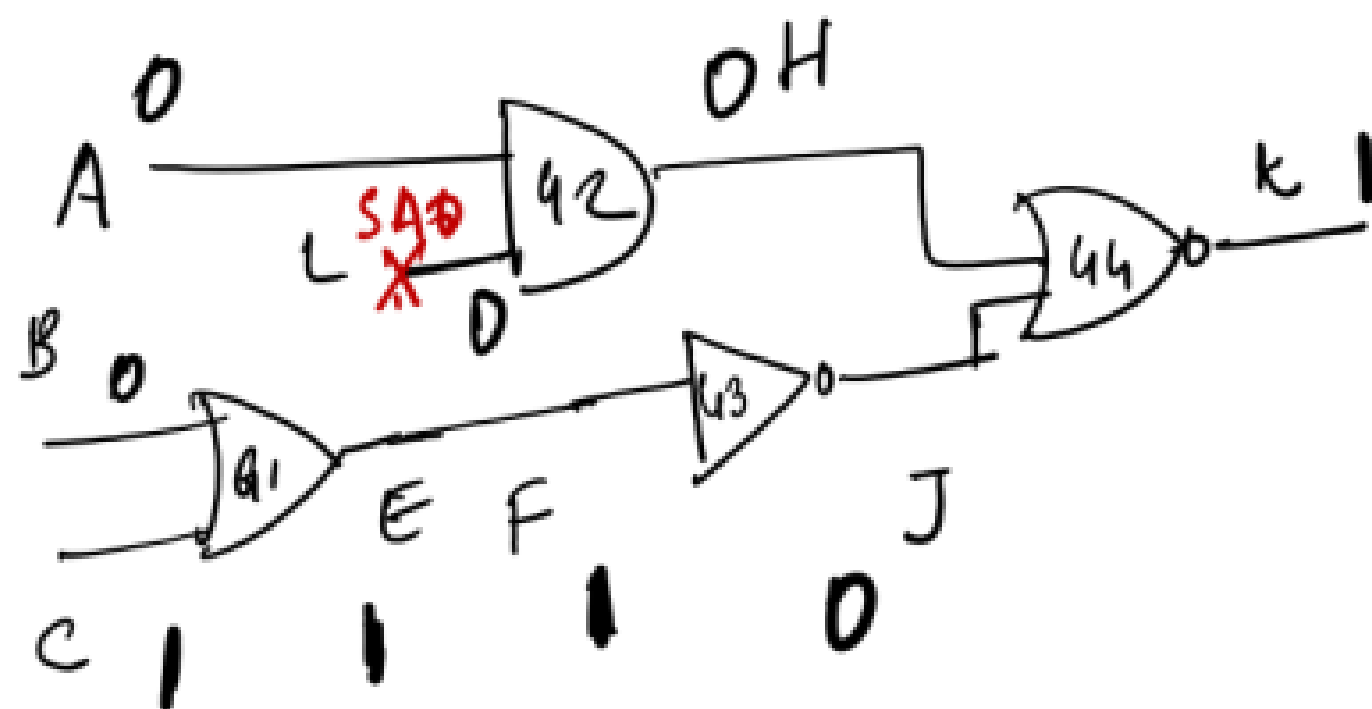
Faulty m/c with ESA0 $\Rightarrow 0$



Faulty m/c with FSA0 $\Rightarrow 0$

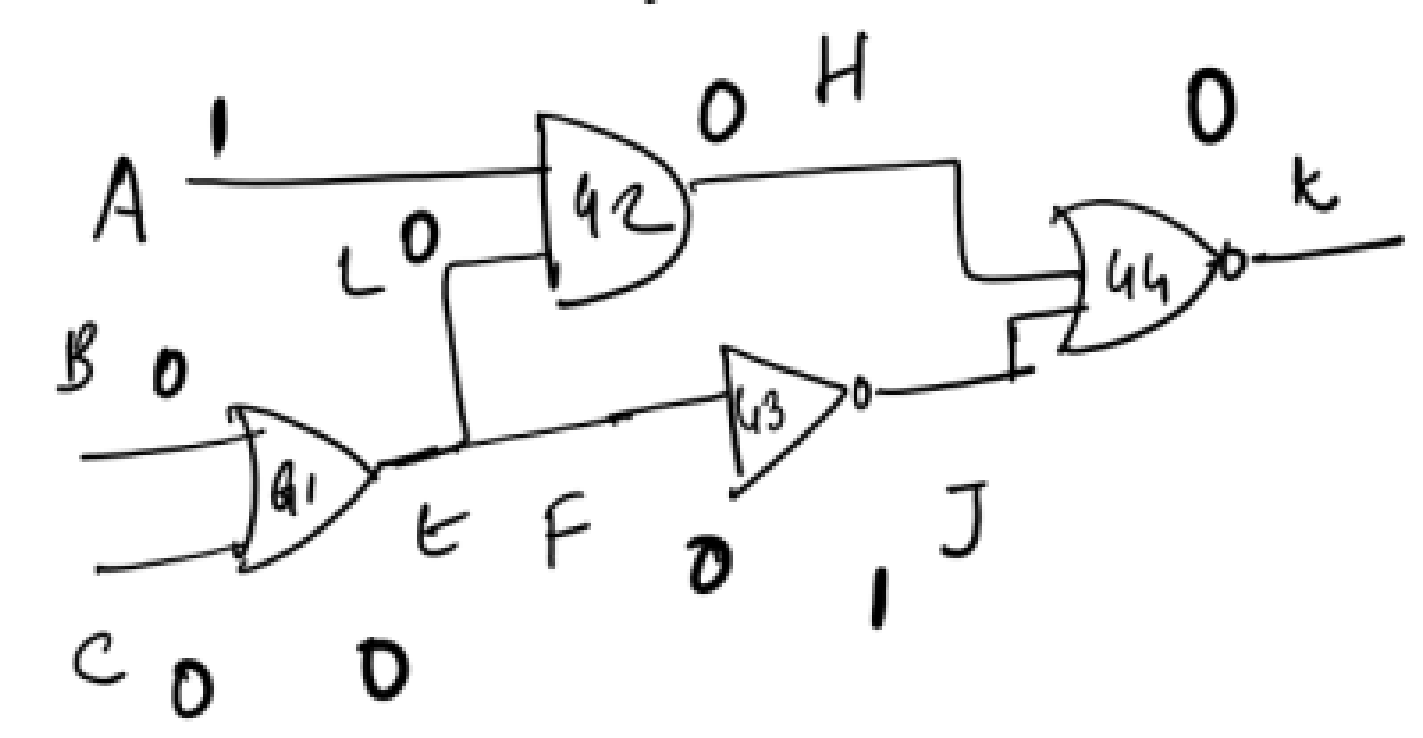


Faulty m/c with LSA0 $\Rightarrow 1$

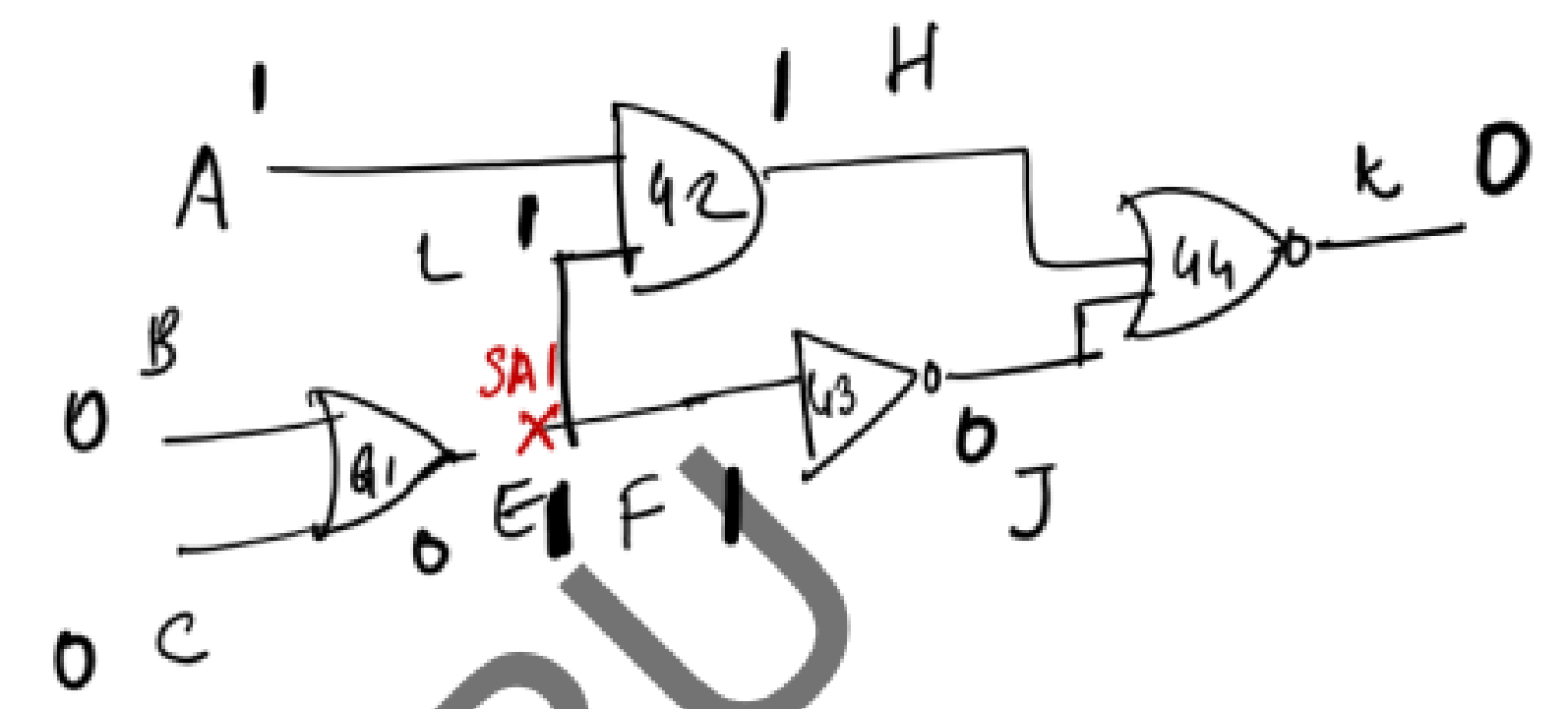


$A=1$
 $B=0$
 $C=0$

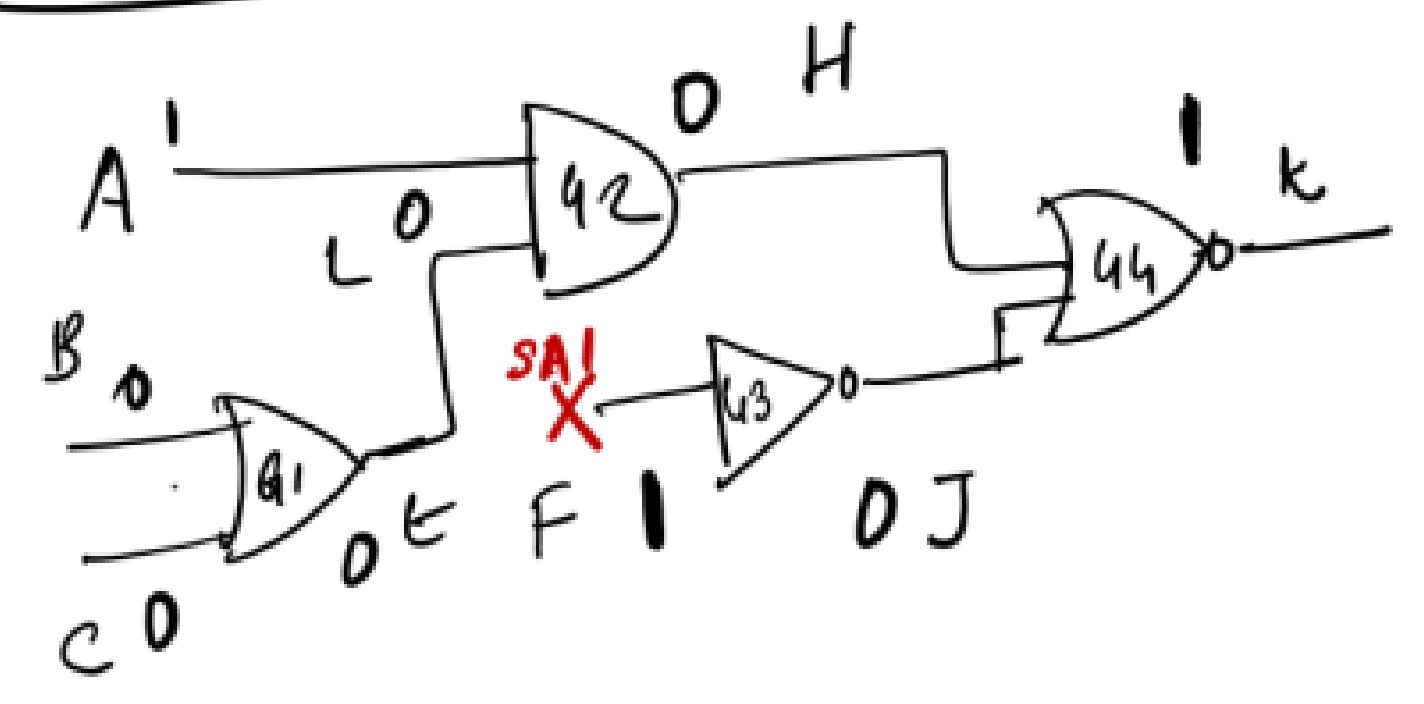
Good m/c $\Rightarrow 0$



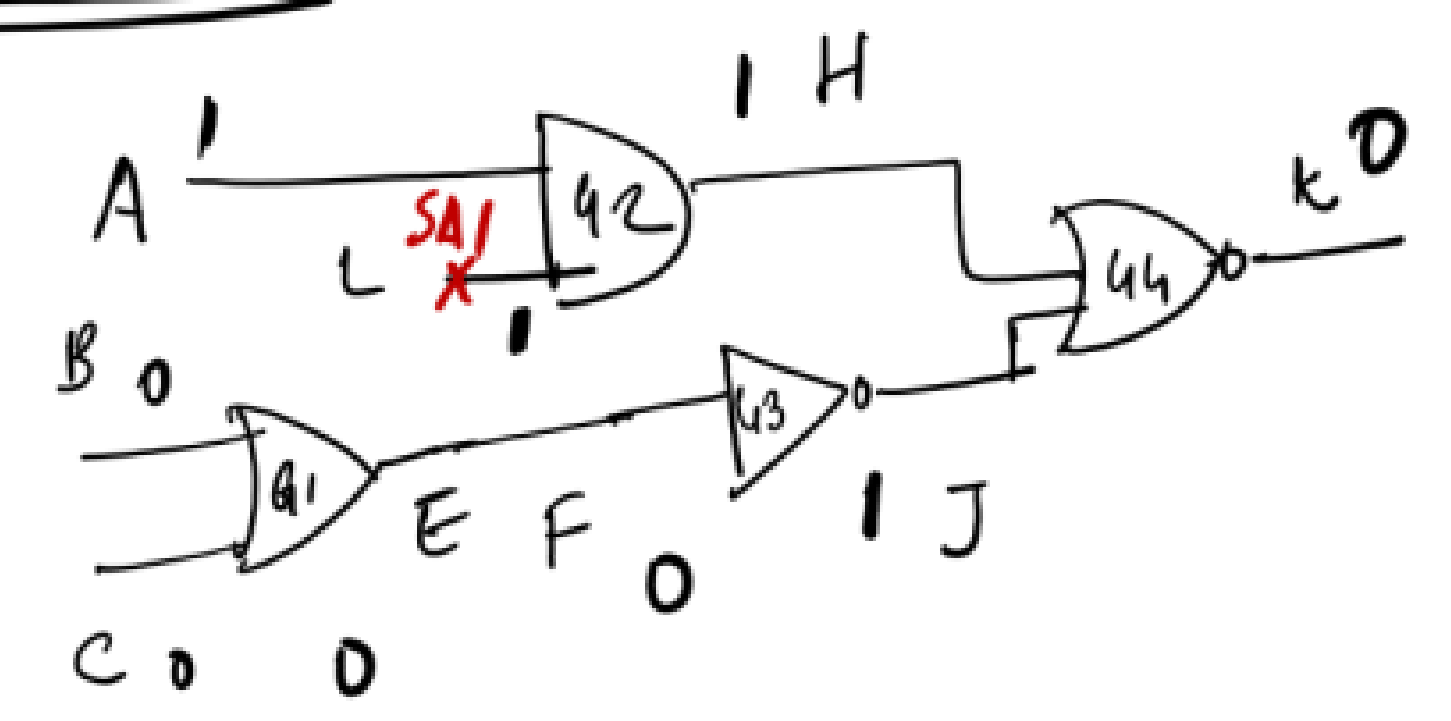
Faulty m/c with ESAI $\Rightarrow 0$



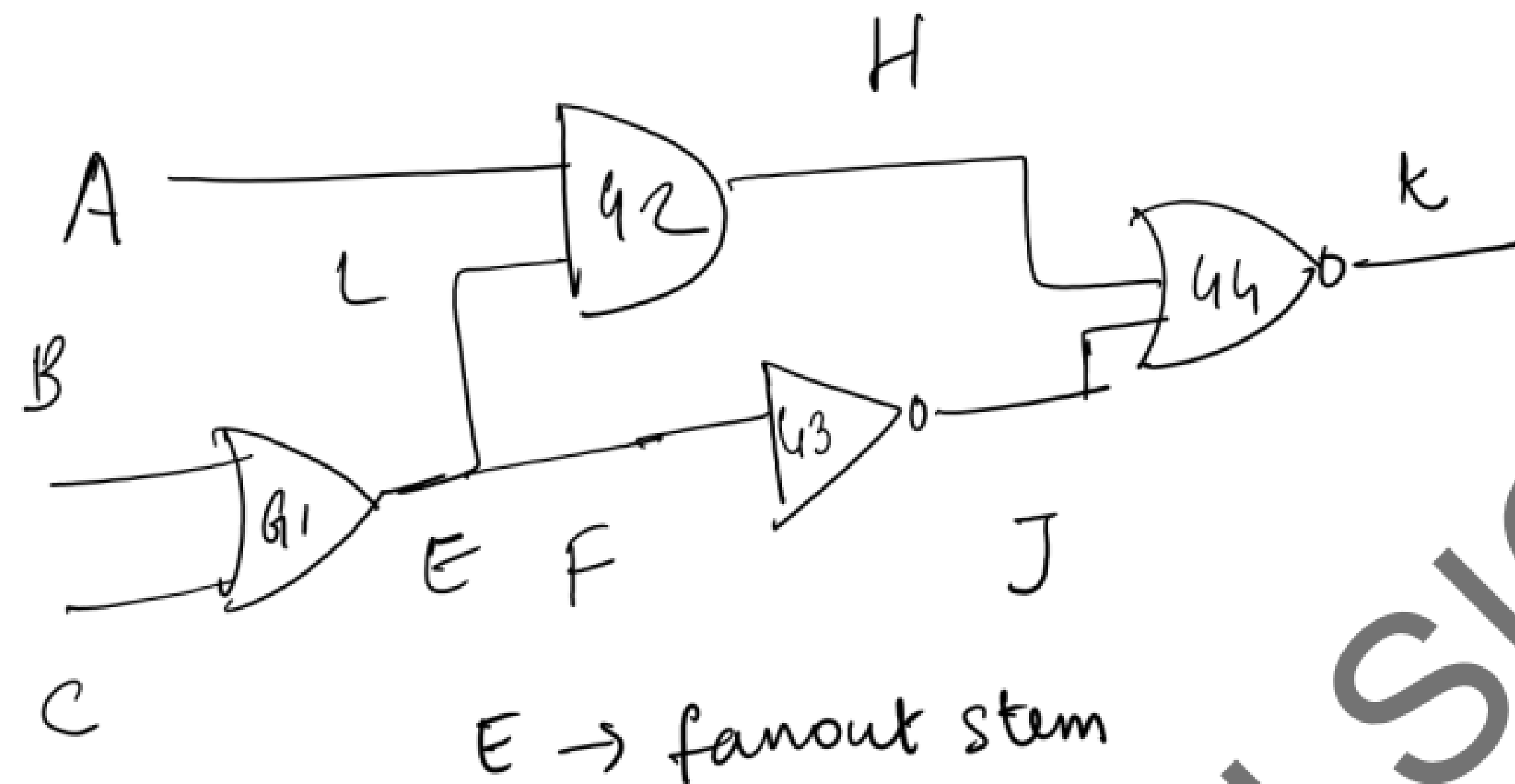
Faulty m/c with FSAI $\Rightarrow 1$



Faulty m/c with LSAI $\Rightarrow 0$



CONCLUSION:-



E → fanout stem

L, F → fanout branches

E, L, F $\Rightarrow$ are o/p's of G1

But it should not be considered

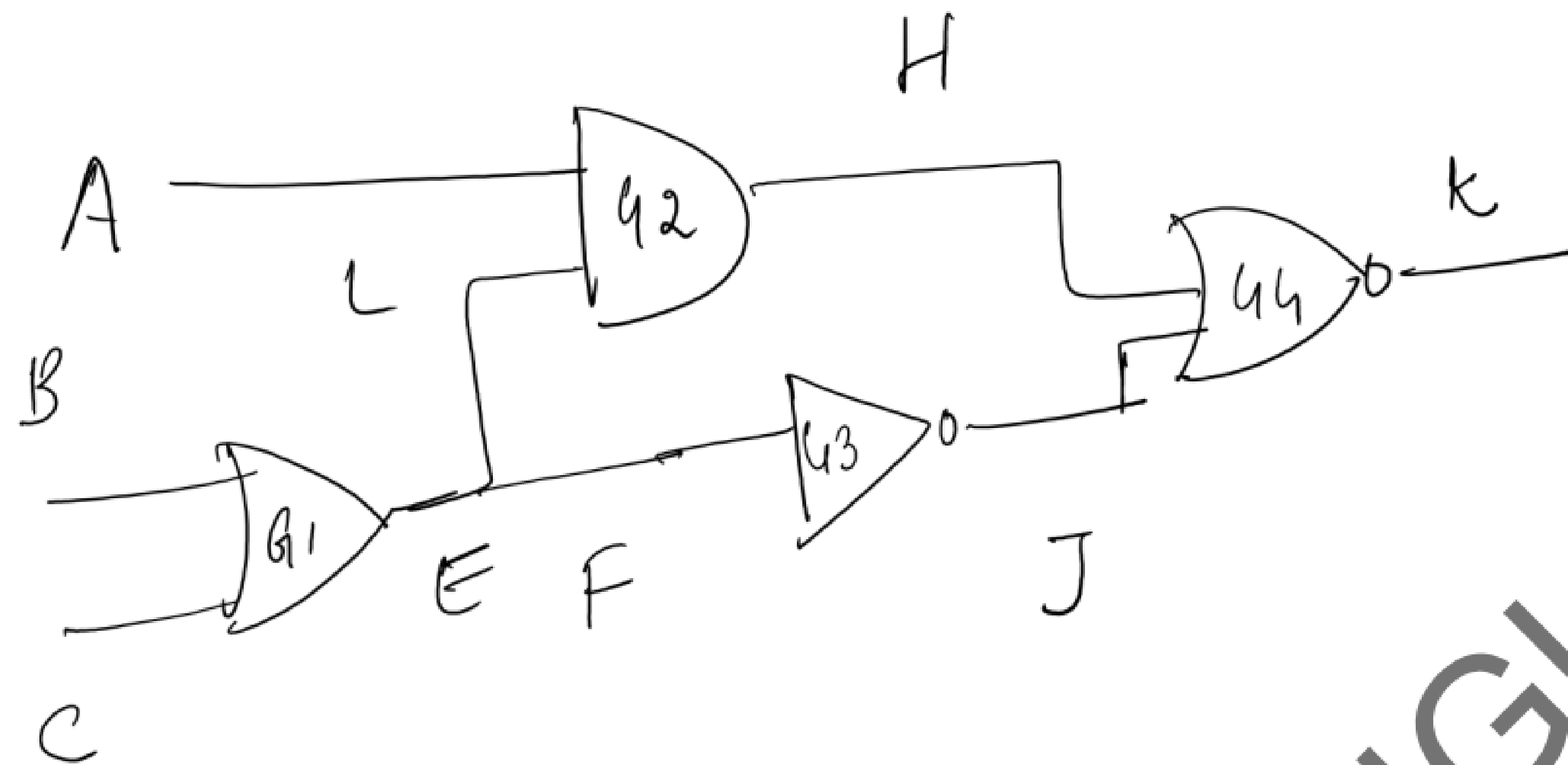
as same.

As illustrated in the previous 2 slides, the faulty response with each of these faults were diff.

$\therefore$ We have to consider SA faults on

E, L, F different.

$\therefore$ SA fault on fanout stem is not equivalent to the SA faults on fanout branches.



How many SA faults are there in this ckt?

⇒ 18 SA faults

ASAO	BSAO	CSAO	ESAO	LSAO	FSAO	HSAO	JSAO	KSAD
ASAI	BSAI	CSAI	ESAI	LSAI	FSAI	HSAI	JSI	KSAI

COVERAGE

It is a measure of percentage of faults that can be detected by the generated test patterns out of the total no. of faults that could exist in a circuit.

$$\text{Fault Coverage} = \frac{\text{Total no. of faults detectable}}{\text{Total faults in a circuit}}$$

Where total faults includes testable faults and untestable faults.

$$\text{Test Coverage} = \frac{\text{Total no. of faults detectable}}{\text{Total faults testable}}$$

Test Coverage will be more compared to fault coverage as in test coverage untestable faults are not included.

We will mainly try to improve Test Coverage because untestable faults will not affect the functionality of the design.

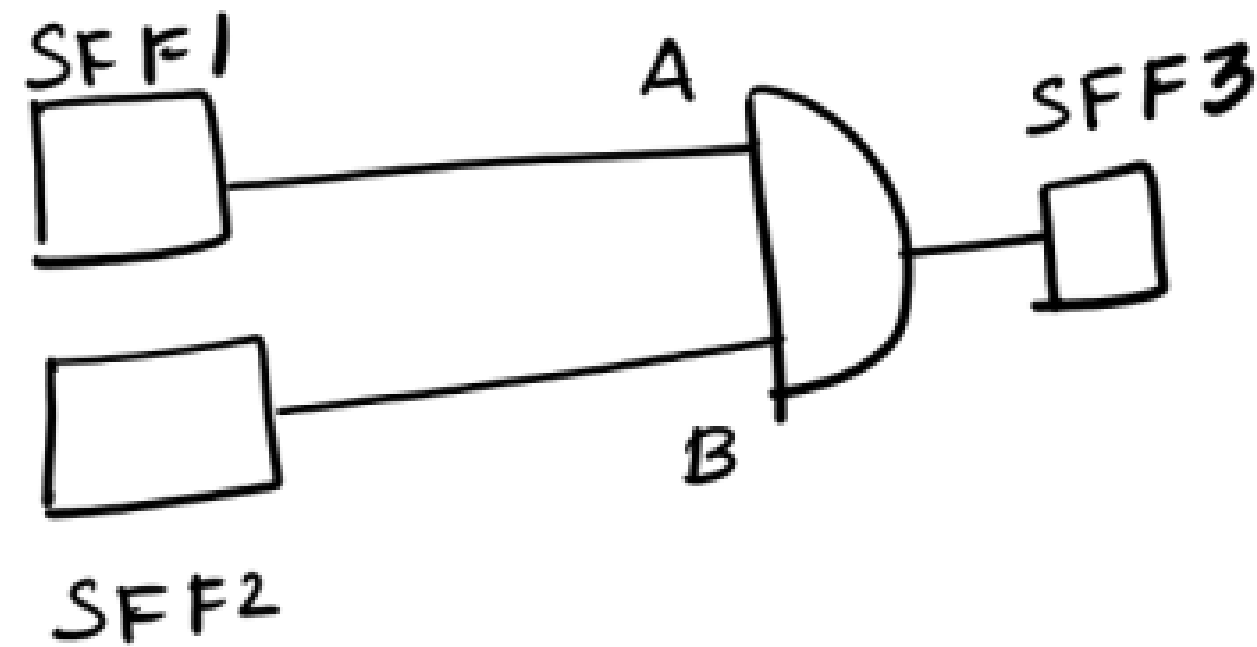
the faults in untestable class, if suppose it occurred after man, and it is not detected, it will not affect the fn. even if

*Dr of test cov is less than the Dr of fault cov.
∴ Test cov is more compared to fault cov.*

COVERAGE :-

It is a measure of percentage of faults that can be detected by the generated test patterns out of the total no. of faults that could exist in a circuit.

Eg:- DESIGN



pattern list :- {01, 10, 11}

Total no. of faults that the tool will inject in this design } = 6
during fault sim

with {01, 10, 11} pattern, total no. of faults that is detected } = 6.
during Fault Sim

$$Cov = \frac{6}{6} \times 100 = 100\%$$

if any of the faulty behaviour as modelled/ mimicked by the tool occurs after man due to a presence of defect, then it will get detected with our test pattern.

Cov = 98%. $\rightarrow$ 98% of the faulty behaviours which is modelled/mimiced by the tool (faults) is detected

$\Downarrow$ (imply)

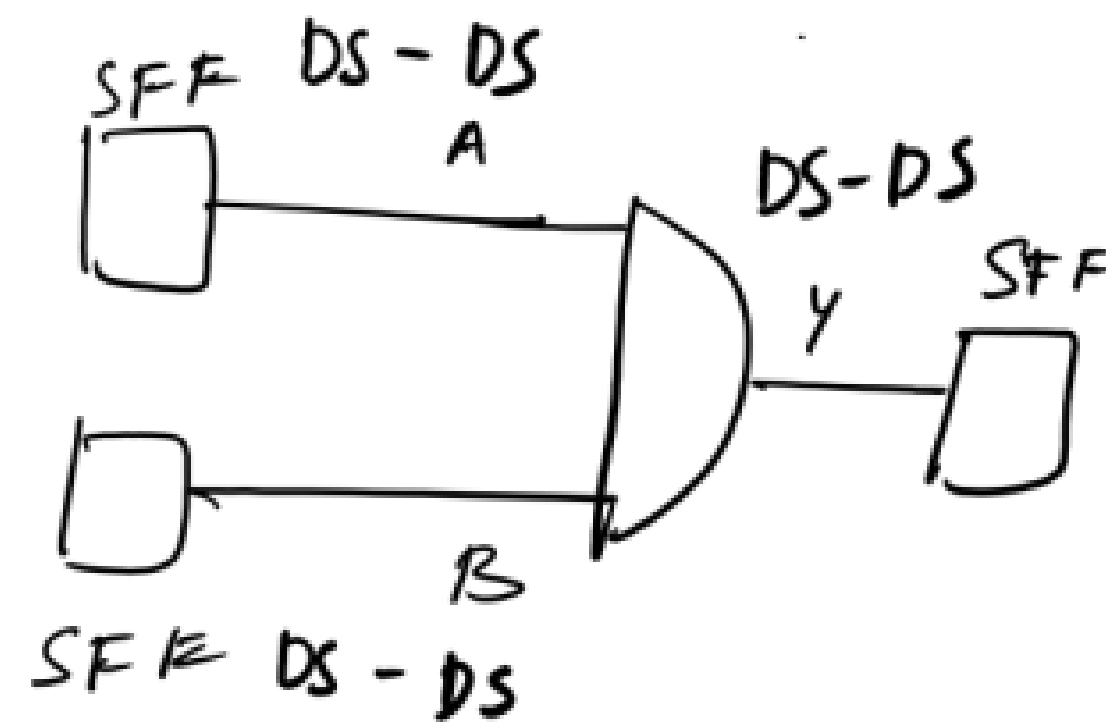
if any of those faulty behaviours as modelled/mimiced by the tool occurs after man due to a presence of defect, then it will get detected with our test pattern.

ROLE OF DFT ENG :-

COVERAGE ANALYSIS :- in the above eg, we need to see why the tool was not able to find a pattern in order to detect those remaining 2% of faults.

COVERAGE IMPROVEMENT :- We need to do something (Eg:- testpoint insertion) so that the tool will be able to find a pattern to detect those remaining 2% of faults.
So that if those faulty behaviours as modelled/mimiced by the tool occurs after man due to a presence of defect, it can get detected.

AND gate



pattern { 01, 10, 11 }

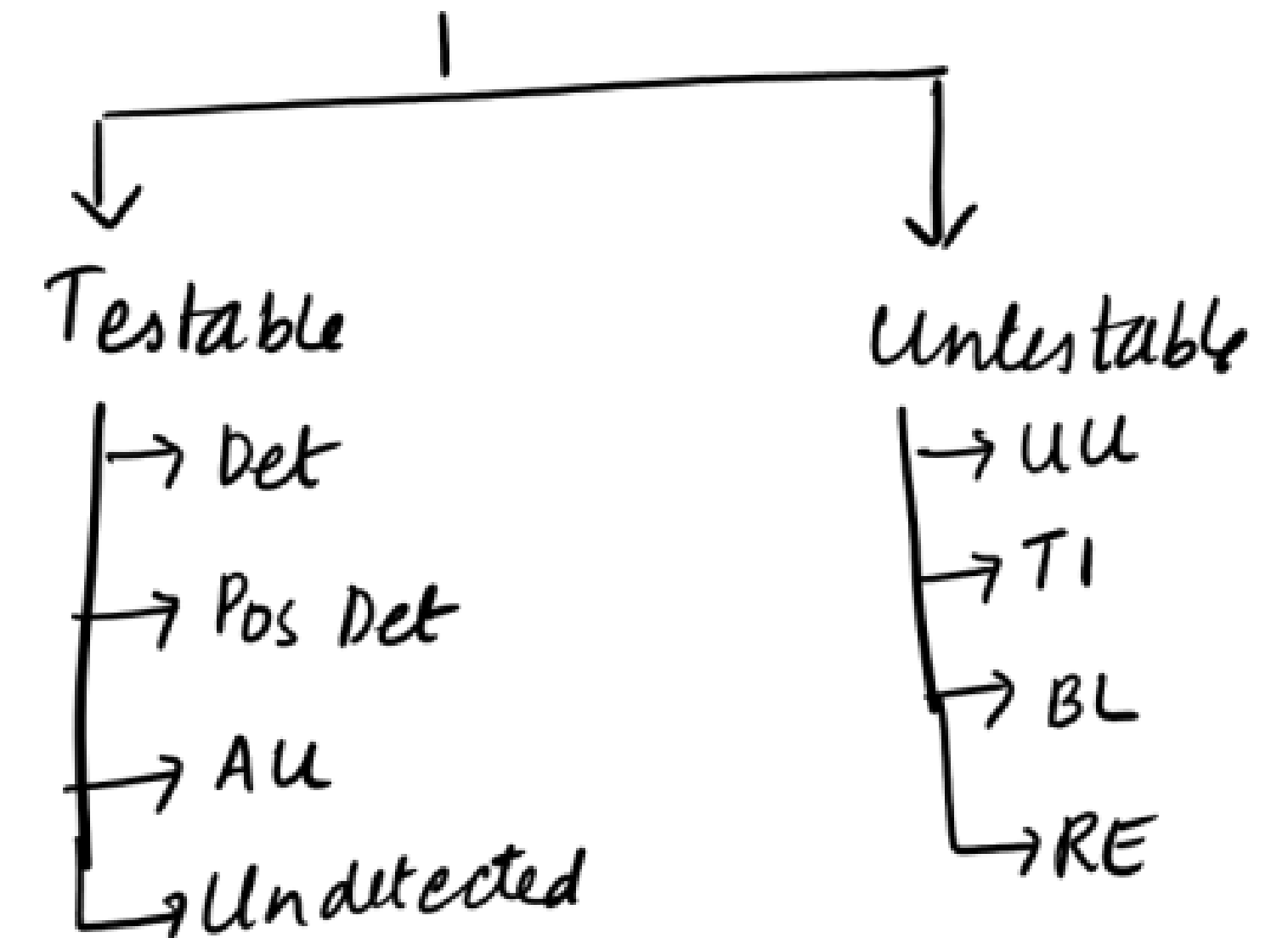
Total no. of faults that the tool will inject in the design = 6

Total no. of faults that is detected during fault sim Fault } = 6.

$$\text{Test cov} = \text{Fault cov} = 100\%$$

No. of faults testable = 6

$$\begin{aligned} \text{Total faults} &= \text{testable faults} + \text{untestable faults} \\ &= 6 + 0 \\ &= 6 \end{aligned}$$

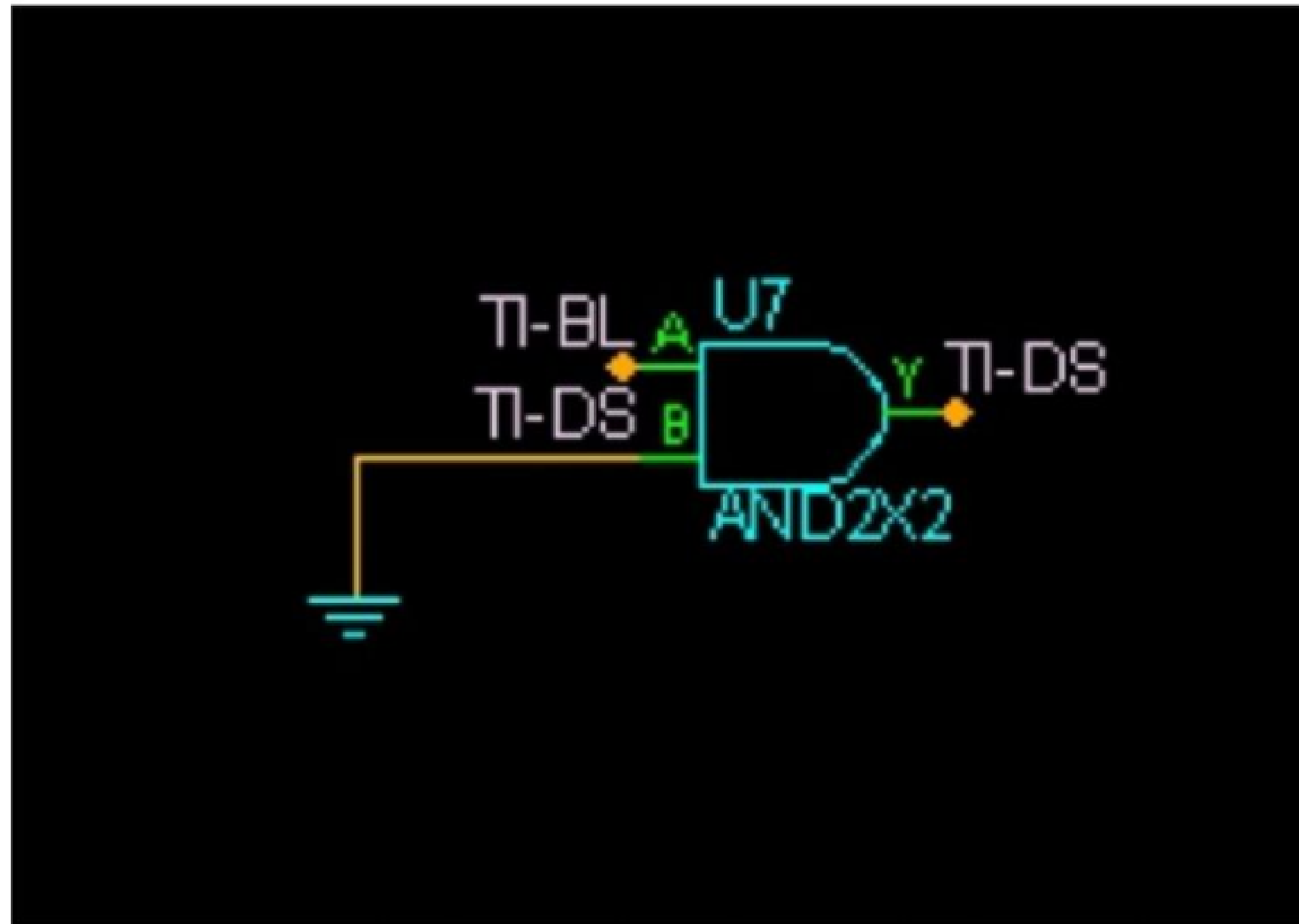


$$\text{Test cov} = \frac{6}{6} \times 100 = 100\%$$

$$\text{Fault cov} = \frac{6}{6+0} \times 100 = 100\%$$

Total no. of faults that the tool will inject in this design = 6 (ASAO, ASAI, BSAO, BSAI, YSAO, YSAI)

Total no. of faults that is detected during Fault sim = 2 (BSAI, YSAI)



TI, BL $\Rightarrow$ Untestable fault class

Total faults = 6

No. of faults testable = 2

No. of faults untestable = 4

$\hookrightarrow$ ASAO (TI)

ASAI (BL)

BSAO (TI)

YSAO (TI)

$$\text{Test cov} = \frac{2}{2} \times 100 = 100\%$$

$$\text{Fault cov} = \frac{2}{6} \times 100 = 33.33\%$$